Practical No. 07

Title of the Assignment: Implement Backtracking Search to Solve the 8-Queens Constraint Satisfaction Problem

Problem statement: Implement backtracking search to solve the 8-Queens Constraint Satisfaction Problem. The program should place eight queens on an 8 × 8 chessboard such that no two queens attack each other.

Objective:

- To understand the concept of Constraint Satisfaction Problems (CSPs).
- To understand the backtracking search technique for solving a CSP.
- To formulate the 8-Queens problem using variables, domains and constraints.
- To implement backtracking search and obtain a valid 8-Queens solution.

Theory:

Constraint Satisfaction Problem (CSP):

A Constraint Satisfaction Problem (CSP) is a problem in which a set of variables must be assigned values from their respective domains while satisfying a set of constraints. A CSP is commonly represented using three components: variables, domains and constraints.

For the 8-Queens problem, one practical representation is to use one variable for each row. The value assigned to a variable represents the column in which the queen is placed.

8-Queens Problem:

The 8-Queens problem requires placing eight queens on an 8 × 8 chessboard so that no two queens can attack each other. A queen can attack another queen if they are in the same row, same column, or same diagonal.

CSP representation:

- Variables: Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, where Qi represents the queen placed in row i.
- Domain: {1, 2, 3, 4, 5, 6, 7, 8}, where the value represents the column number.
- Row constraint: Only one queen is placed in each row because each variable represents one row.
- Column constraint: No two queens can be placed in the same column.
- Diagonal constraint: For any two queens Qi and Qj, $|Qi - Qj| \neq |i - j|$.

Backtracking Search:

Backtracking is a systematic search technique used to solve CSPs. It assigns a value to one variable at a time. If the current assignment violates any constraint, the algorithm rejects that value and tries another value. If no value is possible for the current variable, the algorithm backtracks to the previous variable and changes its assignment.

For the 8-Queens problem, queens are placed row by row. At each row, the algorithm checks whether a column is safe. If a safe column is found, the queen is placed and the algorithm moves to the next row. If no safe column is available, the previous queen is removed and another column is tried.

Input Data / Problem Data:

- Chessboard size: 8 × 8
- Number of queens: 8
- Rows: 1 to 8
- Columns: 1 to 8
- Initial state: No queens placed
- Goal state: Eight queens placed with no pair attacking each other

Constraint Checking:

- Same column: Two queens must not have the same column number.
- Same diagonal: Two queens must not satisfy |column1 − column2| = |row1 − row2|.
- Same row: Automatically avoided because exactly one queen is considered for each row.

Example Solution:

One valid solution can be represented as [1, 5, 8, 6, 3, 7, 2, 4]. This means:

- Q1 → Row 1, Column 1
- Q2 → Row 2, Column 5
- Q3 → Row 3, Column 8
- Q4 → Row 4, Column 6
- Q5 → Row 5, Column 3
- Q6 → Row 6, Column 7
- Q7 → Row 7, Column 2
- Q8 → Row 8, Column 4

Board representation:

Q . . . . . .

. . . . Q . . .

. . . . Q

. . . . . Q . .

. . Q . . . . .

. . . . . . Q .

. Q . . . . . .

. . . Q . . . .

Backtracking Algorithm:

1. Step 1: Start.
2. Step 2: Initialize an empty 8 × 8 chessboard or an empty solution array.
3. Step 3: Start with row 1.
4. Step 4: Try each column from 1 to 8 for the current row.
5. Step 5: Check whether placing a queen at the selected position is safe.
6. Step 6: If the position is safe, place the queen and move to the next row.
7. Step 7: If all eight rows are successfully assigned, a solution is found.
8. Step 8: If the position is not safe, try the next column.
9. Step 9: If no column is safe for the current row, remove the queen from the previous row and backtrack.
10. Step 10: Try another column for the previous row.
11. Step 11: Continue the process until a solution is found or all possibilities are exhausted.
12. Step 12: Display the solution and chessboard.
13. Step 13: Stop.

Pseudo Code:

```
BACKTRACK(row)
   if row > 8:
      return TRUE
   for col = 1 to 8:
      if SAFE(row, col):
         place queen at (row, col)
         if BACKTRACK(row + 1) == TRUE:
            return TRUE
         remove queen from (row, col)
   return FALSE
```

SAFE(row, col):

- Check whether another queen is already in the same column.
- Check whether another queen is on the same diagonal.
- Return TRUE if the position does not violate any constraint.
- Otherwise, return FALSE.

Illustrative Search / Backtracking:

The search begins from Row 1. If a selected column later causes a conflict, the algorithm goes back to the previous row, removes the previously placed queen, and tries the next available column. For example, a partial assignment such as [1, 5, 8] is extended only

when the next position is safe. If no safe position is available at a later row, the algorithm backtracks and changes an earlier queen's position. This continues until a complete valid arrangement is obtained.

Advantages and Limitations of Backtracking:

Advantages:

- Systematic method for solving constraint satisfaction problems.
- Reduces the search by rejecting invalid partial assignments early.
- Simple to understand and implement using recursion.
- Can be applied to N-Queens, Sudoku, graph coloring and other CSPs.

Limitations:

- The worst-case search space can be large.
- Performance depends on the order in which variables and values are selected.
- For larger N, basic backtracking may require additional heuristics such as forward checking or MRV.

Conclusion:

The 8-Queens problem was formulated as a Constraint Satisfaction Problem using variables, domains and constraints. Backtracking search was used to place queens row by row while checking column and diagonal constraints. Whenever a partial assignment became invalid, the algorithm backtracked and tried an alternative assignment. A valid arrangement of eight non-attacking queens was obtained.

Oral Questions:

1. What is a Constraint Satisfaction Problem (CSP)?
14. What are the variables and domains in the 8-Queens problem?
15. What are the constraints in the 8-Queens problem?
16. What is backtracking search?
17. Why is backtracking required in the 8-Queens problem?
18. How do you check whether a queen can be placed safely?
19. What is the diagonal constraint for two queens?
20. What is the difference between backtracking and brute-force search?
21. Why is recursion commonly used in backtracking?
22. What are some applications of backtracking in AI?